

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

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Website GitHub LinkedIn Google Scholar

Profile

Applied ML researcher and engineer with a Ph.D. in Computer Science and financial AI experience at Nasdaq. First-author publications in **IEEE TNNLS** and **TVCG**; **NeurIPS** research integrated into **Hugging Face PEFT**. Combines generative vision research with ML platform integration and cloud deployment.

Technical Skills

Core: Python, PyTorch, SQL, Git, Bash, C++, CUDA; experimental design and feature engineering.

ML & GenAI: Latent diffusion, Transformers, multimodal AI, PEFT/LoRA, Hugging Face, RAG, agentic tools.

ML systems: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Professional Experience

Senior Data Scientist, AI/Analytics – Nasdaq

Mar. 2025 – Present

St. John's, NL, Canada

- Deployed an **existing UI/backend prototype** of a cross-team ML training platform into an internal workspace; configured network connectivity and IAM roles for internal access.
- Implemented **PostgreSQL-backed user-state persistence** and deployed **FastAPI** services to **AWS Lambda using Mangum**, supporting the platform's UI/backend integration.
- Designed cross-scale ensemble feature selection, removing **77.2% of input features** while retaining competitive performance on the CFML dataset; presented the work to **60+ attendees**.
- Built a unified Python workflow for **SageMaker Processing, Training, and Hyperparameter Tuning**; automated dependency packaging and linked jobs using **SageMaker Pipelines**.
- Developed a **Bedrock-based feature-formula validator** to assess logical consistency and flag hard-coded feature-engineering errors.
- Built a **RAG-based incident-response assistant** to retrieve troubleshooting context for internal engineering workflows and on-call alerts.

AI Algorithm Intern – Nasdaq Verafin

May 2024 – Jan. 2025

St. John's, NL, Canada

- Designed diffusion-based document image restoration and inpainting methods for degraded cheque images; implemented and evaluated the approach in Python and PyTorch.
- Developed an image-verification approach using patch-level Gaussian latent modeling, and presented research progress to cross-functional colleagues.

AI Intern – Beaufort Solutions

Jul. 2022 – Nov. 2022

St. John's, NL, Canada

- Designed a data-collection pipeline and few-shot image-theme recognition using vision-language models for personalized photo-book curation; contributed to TEG, published in Journal of Electronic Imaging (2024).

AI Algorithm Intern – Zhongkong Hanlian

Dec. 2017 – Aug. 2018

Hangzhou, China

- Built vehicle-logo recognition and annotation tools using deep networks, Python, C++, and Java; supported deployment for real-time recognition at community security gates.

Selected Research Publications

UltraDiffEdit – High-Resolution Generative Editing

First author; IEEE TNNLS 2026

- Developed tuning-free editing with multi-patch encoding, global-local denoising, and progressive sampling; demonstrated **8K editing on a single RTX 3090**. Code; video.

FACEMUG – Multimodal Local Facial Editing

First author; IEEE TVCG 2024

- Developed multimodal feature fusion and self-supervised latent warping for local edits guided by text, sketches, semantic/color maps, exemplars, and attributes, preserving unedited regions.

AdaMSS – Parameter-Efficient Adaptation

Co-author; NeurIPS 2025

- Co-authored adaptive multi-subspace fine-tuning for language and vision models; the method is **integrated into Hugging Face PEFT**. Code; PEFT integration.

DocPure – Unified Document Restoration

Co-first author; IEEE TCSVT 2026

- Co-authored prompt-free document restoration using degradation-aware structure extraction and structure-guided wavelet modulation across multiple degradations. Code.

GRIG – Data-Efficient Image Inpainting

First author; CVM 2025

- Developed iterative residual inpainting using CNN features, Transformer reasoning, and forgery-patch adversarial training for limited-data settings. Code.

Visual Style Prompt Learning

First author; Pattern Recognition 2025

- Developed diffusion-generated visual prompts and style-modulated aggregation for blind face restoration. Code.

Full publication record: [Google Scholar](#) and [publication portfolio](#).

AI Engineering Prototypes & Deployment

Local Agentic Code Editor

Nasdaq internal prototype; 2026

- Prototyped a local AI code editor with an **agentic loop** that selects between code writing, code review, and shell-tool execution based on the current task.
- Integrated **language-model APIs with local tool execution** to support code generation, debugging, and iterative review.

AI Headshot Generation Deployment

Model integration & deployment

- Implemented model invocation and application deployment for AI-generated headshots.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada

2021–2025

M.Sc., Computer Software and Theory – Wenzhou University, China

2018–2021

B.Sc., Digital Media Technology – Communication University of Zhejiang, China

2014–2018

Selected Recognition & Service

Outstanding Research Award (Ph.D.), Memorial University, 2023; four granted Chinese invention patents.

Mentored **10 students** on generative vision, image restoration, recognition, and applied ML research.

Invited reviewer for **IEEE TPAMI, TIP, TMM, TCSVT**, Pattern Recognition, Knowledge-Based Systems, Neurocomputing, Applied Soft Computing, and ECCV 2026.

Invited speaker, IEEE Newfoundland and Labrador Section AGM, 2026; guest lecturer on deep generative models, Memorial University, 2025.